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(1) Research Program/Project/Study Title: <u>Healthcare Information System for Roxas City's Pandemic, Epidemic, Preparation and Surveillance (PEPS)</u>	
Proponent(s) Name:	Balingit, R., Nava M., Amanguiton Z., Vergabera R., Dionio R., Almonguera C., Begas J.M., Anoché J.M., Bartolo K.
Academic Rank:	Associate Professor II
Co-Authors (with academic rank):	Nava M.(Asst. Prof II),Amanguiton Z. (Asst. Prof IV), Vergabera R. (Instructor III), Dionio R. (Professor I), Almonguera C. (Assoc. Prof. 1), Begas J.M., Anoché J.M., Bartolo K.
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Program or Project has a component (if no proceed to 2)? ☐ Yes ☐ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	City of Roxas						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☒ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☒ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☒ Pilot Testing	☒ Applied	☒ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☒ Pilot Testing								
☒ Applied	☒ Tech. Promotion/Commercialization								

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(7) Sector/Commodity

Improving Healthcare Services

(8) Discipline/Field of Specialization

Healthcare Data, Computing and Mobile Application, Databases, Information System, Data Governance, Telehealth, IT Infrastructure

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users, and the country.)

Health information is the data related to a person's medical history, including symptoms, diagnoses, procedures, and outcomes (ahima.org, 2024). The City Health of Roxas manages the health information data of the forty-seven Barangay Health Units (BHUs) that serve the immediate healthcare needs of the poor and indigent patients of the community. However, this health information data is mostly written on paper (or forms) and stored in folders that can be misplaced and/or destroyed by unprecedented calamities (i.e. fire, typhoon, floods). The accountability of Health Information data is not available due to no formal policies as to who are the health data stewards, or no Health Information Data Governance guidelines are made.

The Healthcare Information System (HIS) can streamline the collection of storage of data and support efforts that improve local civil registration and vital statistics, especially in underserved communities. The developed application tool at the front end greatly increases the health service delivery network effort in the local communities. Furthermore, the HIS implements the health interoperability standards that promote uniformity and consistency of health information data across local and national collections, conform to a set of health information data unique codes used for software development, and can execute seamless exchange of various health resources across the country.

Digitalizing and the collection of more structured standardized health **information** data, together with the developed and implemented informatics resources/system in place will help support the policies and decisions of the City Health Local Government Unit. With this health information data in place, the City Health LGU can perform health data analytics and be able to visualize the diseases through AI solutions (e.g. Health Map) that can improve decision-making and policies for prevention, preparedness, and response to such disease outbreaks.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

The research project aims to design, develop, and implement a more accessible and strengthened public health system that leverages the City Health of Roxas's responsible solutions (through AI) to improve prevention, preparedness, and response to disease outbreaks. To perform such prevention, preparedness, and response, the existing health information data (if there is any) will be curated, migrated, and/or fed to the HIS. More health information data are needed to be able to perform data analytics, visualization, and better responses to formulating policy decisions to counter, minimize, and avoid disease outbreaks in Roxas City. Additionally, the application tool is developed and installed in certain Android (or IOS) operated tablet devices with mobile data Internet connectivity that will be used by a trained Barangay Health Worker (BHW) that collects the preliminary health information data from a patient and Medical Doctor (or City Health Doctors) that conducts a telehealth examination through a video call. Figure 1 shows the overall architecture.

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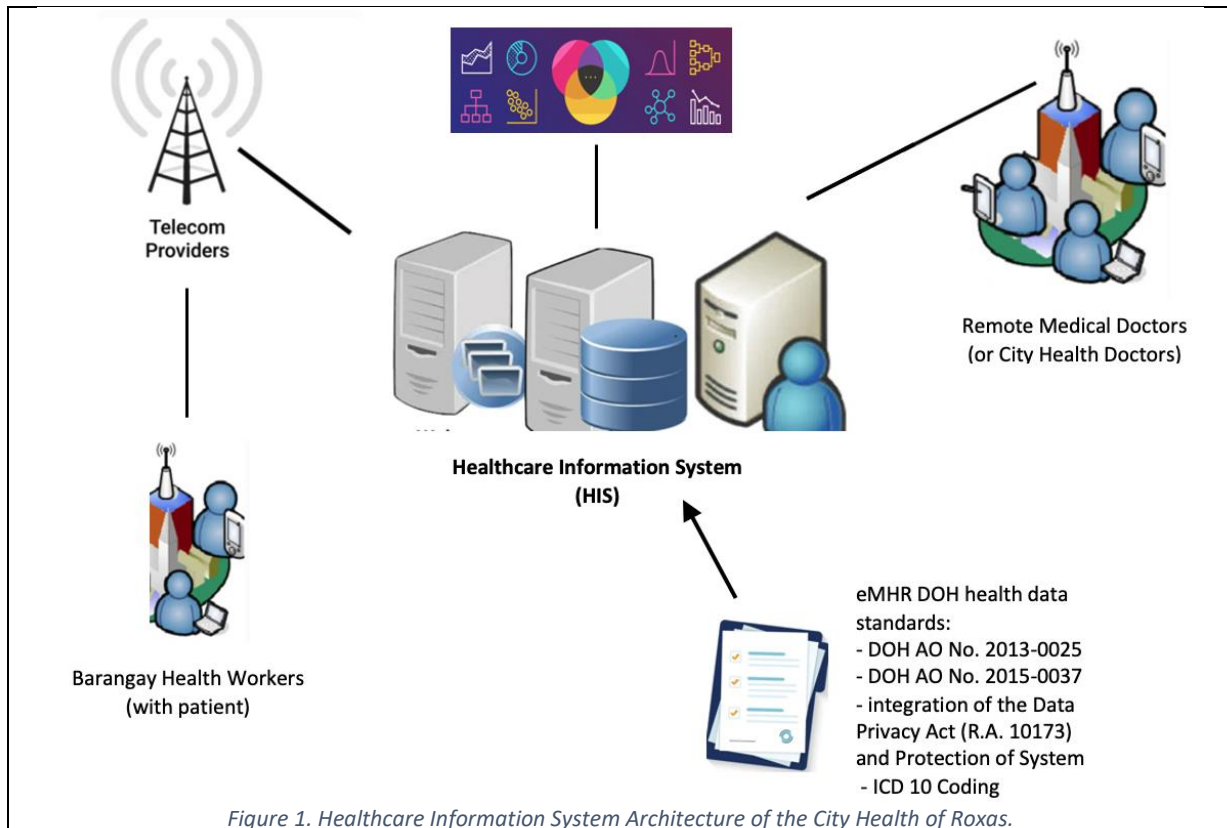


Figure 1. Healthcare Information System Architecture of the City Health of Roxas.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

The general objective is to design and develop a more accessible and strengthen public health system that leverage City Health of Roxas responsible solutions (through AI) to improve prevention, preparedness, and response to disease outbreaks.

- To design/develop an HIS infrastructure that stores collected health information data of patients of the 47 barangay health centers in Roxas City.
- To develop a standardized patient eMHR (electronic Medical Health Record) that aligns with DOH's national implementation of health data standards (e.g. DOH AO No. 2013-0025, DOH AO No. 2015-0037).
- To design/develop an application tool that improves the quality of patient care while protecting patient's collected health information data.
- To develop/implement an equitable Health Information Data Governance for responsible data stewardship and with ethical/responsible behaviors for sourcing, sharing, and implementing data that will cause no harm.
- To develop a training plan for the Barangay Health Workers and City Health Doctors on the developed application tool (or telehealth tool).
- To conduct training for both Barangay Health Workers and City Health Doctors.

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

The ATIPAN Project of the USA-CFI bring digital health to 10 underserved and vulnerable indigenous and rural low-income communities in Western Visayas. ATIPAN Project provides a free medical teleconsultation, and with technology training to 23 community health workers. The project collected health information data through 7108 consultations; determine that 62.55% are women and girls, and 61.10% are children, adolescent, and elderly. [1]

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In the health sector, AI methods are used in studying patient's data and foretelling upcoming incidence and creating smart interface for communication with the patient improvement in his/her commitment with the treatment plan [3].

Artificial Intelligence for Pandemic Epidemic Preparedness (AI4PEP) is an initiative that will address existing knowledge and practice gaps in the Global South (where the Philippines is part of) by establishing a multi-regional network to deepen the understanding of how responsible AI solutions can improve public health preparedness and response [2].

To keep the project running, manpower is highly essential. The integration of Data Privacy (RA 10173) and Protection System and learning ICD10 serves as a starting point in training the stakeholders and the research project team as well. The conduct of capability-building activities on e-health systems, public key infrastructure, enterprise architecture server repository tools, service-oriented architecture applications, and terminology services strengthens the users' capacity to manage these technologies with minimal support. [4]

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

The project will adopt the Framework for clinical public health needs of the Global South presented by AI4PEP. Note that Philippines belongs to the so-called Global South. It contains three shells; the inner shell ("how") contains the set of ethical and legal rules and codes that should be designed in such a way that they are responsible (in incorporating policy and regulations), locally relevant for communities, and explainable to society at large. Moreover, they should be applied and embedded along all the processes of AI solutions. The medium shell ("what") presents the processes that should be implemented in an iterative fashion (step 1: data collection; step 2: design and development; step 3: deployment; step 4: performance; and step 5: monitoring). [2]

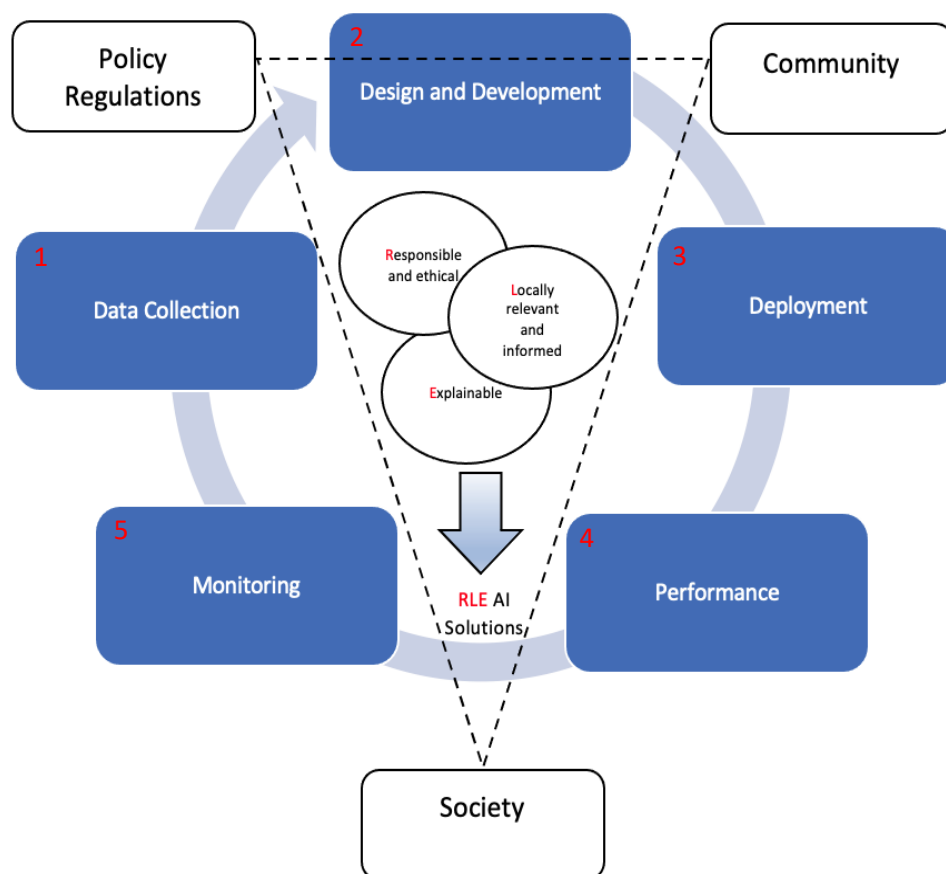


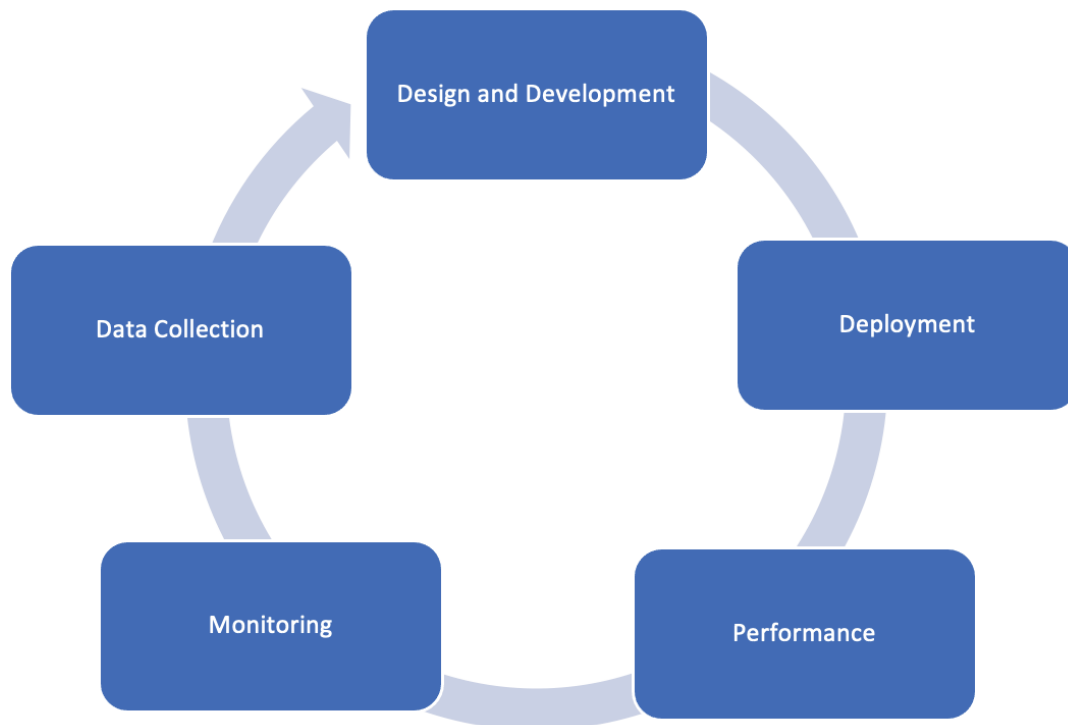
Figure 2. The AI4PEP Framework.

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Responsible (accountable, auditable, compliant, ethical, respectful, safe, secure)
Explainable (equitable, fair, interpretable, reliable, reproducible, transparent, trustworthy, unbiased)
Local (autonomous, caring, connecting, decolonized, human and community-centered, inclusive, intentional, intersectional, just, practical, protecting, process-based, sustainable)

(13) Methodology (Details such as a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

The method that will be used in this research project is the use of a software/system development iterative process. The first step is meeting of the target stakeholders, understanding their present needs, observation, data collection, and other requirement elicitation. The elicited data will be analyzed to determine the needs that are outlined during the iterative meetings with the client. The next step is to execute the design and development wherein the team will be using the “Agile Methodology” iteratively until the needs of the client are met. Then it follows the deployment, evaluation of performance of the system, and monitoring. Note that there is no perfect system, thus, the method is iterated several times to successfully meet the client’s needs. During the second iteration, the team will be conducting end-user training and others (such as statistical analysis / analytics of health information data, visualization, etc.).



Agile Methodology for app Development

The Agile Methodology for app development is a continuous cycle of six steps:

- 1 Brainstorm** (Pink segment)
- 2 Design** (Light blue segment)
- 3 Development** (Pink segment)
- 4 Quality Analysis** (Light blue segment)
- 5 Feedback** (Pink segment)
- 6 ACCEPT Yes** (Light blue segment)

Supporting activities and roles include:

- Identify defects & Resolve bugs** (Linked to Quality Analysis)
- Build Application & Iteration** (Linked to Development)
- Design Application & Prototype** (Linked to Design)
- Requirement analysis** (Linked to Brainstorm)
- Continual Engagement** (Linked to Feedback)
- Client, Developer, User** (Stakeholders involved in the process)
- App Delivered to Client** (Final outcome after the 'ACCEPT Yes' step)

The projected timeline to complete the research project estimated between 3 – 5 years. Thus, the proponents identified the following Phases of activities to accomplish the research objectives:

- [illegible]

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Patient Health Information data												
2. Analysis of data gathered from observation / Learning the ICD 10 Coding / Database Design of the HIS / Digitization of data / eMHR Design												
3. Database Design / eMHR implementation/Coding of HIS (both back-end/front-end) / design of e-MHR template for existing health informatin data for migration to the HIS database / System Development												
4. System Development / Coding / Unit Testing / Pilot Testing with beneficiary / Deployment / Monitoring												

(15) Expected Output (Indicate the specific products, processes, or services which the project is expected to produce; quantify when possible)

Expected Deliverable outputs include:

- Healthcare Information System (w/ Database)
- Mobile/Tablet Application Tool/web-based applications tools to collect anonymized health information data.
- Training Plan Document for BHWs and City Health Doctors
- HIS System and Application Tool Manuals
- Data Dashboard and portals include AI powered tool to analyze/visualize health information data.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes, or services to be transferred as extension activities)

Below are the technology and services to the targeted beneficiaries:

- Healthcare Information System (w/ Database)

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- Mobile/Tablet Application Tool/web-based applications tools to collect anonymized health information data.
- Training Plan Document for BHWs and City Health Doctors
- HIS System and Application Tool Manuals
- *Possibly a Tablet Device with Application Tool installed in it

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

- City Health of Roxas LGU (Medical Doctors, Personnel)
- Barangay Health Workers (of the 47 Barangays of Roxas City)
- Community

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
R. Balingit	Lead / Developer	
M. Nava	Analyst / Developer	
Z. Amanguiton	Analyst / Developer	
R. Vergabera	Analyst / Developer	
R. Dionio	Analyst / Communication to LGUs	
C. Almonguera	Programmer	
J.M. Besa	Programmer	
J.M. Anoché	Programmer	
K. Bartolo	Documentation/Communication to BHWs	

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)		
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)	1 Lot	200,000.00
- Transportation allowance	1 Lot	20,000.00
- Meal allowance during data gathering	10	10,000.00
- Meals during regular meetings with team and with clients	1 lot	20,000.00
- Miscellaneous (for incidental expenses)	1 lot	50,000.00
- Printing		
C. CAPITAL OUTLAY		
- Equipment (1-server, 5-laptops, 2-Tablets)		
- Software (site license, OS)		
- Others	1 lot	2,000,000.00
Total (estimated budget)		2,300,000.00

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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- [1] De Castro R., et. al, <https://www.usacfi.net/atipan-project.html>, 2023)
- [2] AI4PEP, <https://ai4pep.org/about/>, 2023)
- [3] El-Rashidy N, El-Sappagh S, Islam S, M El-Bakry H, Abdelrazek S (2021) Mobile health in remote patient monitoring for chronic diseases: principles, trends, and challenges. *Diagnostics* 11(4):607
- [4] Rosales N.B., <https://nbrosales.wordpress.com/2018/09/01/six-challenges-to-health-informatics-and-their-current-status-in-the-philippines/> , 2018)

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

Balingit R.P.

- Assoc. Professor II, Dept. of Computer Science Dayao Satellite College Capiz State University (Jan 2024 – present)
- Associate Member, National Research Council of the Philippines (July 2023 – present)
- Publications: <https://scholar.google.com/citations?hl=en&user=5go2OgAAAAAJ>

Nava M.P.

- Chairperson, Department of Computer Science, Dayao Satellite College Capiz State University

Amanguiton Z.

- Research Coordinator, Dept. of Computer Science, Dayao Satellite College Capiz State University

Vergabera R.

- Instructor III, Dept. of Computer Science, Dayao Satellite College Capiz State University

Dionio R.

- Director, IQAO Capiz State University

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Anoche J.M.

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Bartolo K.

- Faculty, General Education of Dept. of Computer Science, Dayao Satellite College Capiz State University